A Domain-Adaptive Few-Shot SAR Ship Detection Algorithm Driven by the Latent Similarity Between Optical and SAR Images

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***Abstract*— Detecting ships in synthetic aperture radar (SAR) images poses a formidable challenge, primarily attributed to limited observation samples and complex environments. To address this problem, driven by latent similarity between optical and SAR images, we propose a domain-adaptive few-shot detection algorithm for SAR ship detection [single shot multibox detector (SSD)]. The algorithm requires only a few training samples of SAR images and effectively combines them with rich optical images to utilize domain information. First, we develop an efficient plug-and-play distance metric function. This function accurately measures the distances between features from the optical domain and the SAR domain. Second, we design a lossy branching mechanism to effectively utilize SAR domain knowledge. This branching mechanism is driven by the observed latent similarity in domain knowledge distribution between optical and SAR images. In addition, we introduce a dual-stream branching feature alignment extraction network with weight sharing. This network architecture enables better knowledge extraction and sharing between optical and SAR domains. To evaluate our method, we conducted experiments on a newly created dataset, DIOR2SSDD, which is designed for few-shot SAR image ship detections across optical and SAR domains. The experimental results show that under three-, five-, and ten-shot settings, the mean average precision (mAP) of our method can reach 59.2%, 61.2%, and 64.6%, and with only 10% SAR training data, the mAP can reach 89.3%. It indicates that our method can effectively transfer domain knowledge and achieve excellent ship detection performance in SAR images.**

***Index Terms*— Distribution distance metrics, domain adaptation, domain knowledge, ship detection, synthetic aperture radar (SAR) images.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an essential remote sensing tool, offering high spatial resolution and observation capabilities. It operates in all weather conditions, making it invaluable for monitoring land and ocean surfaces [1], [2], [3], [4]. Fig. 1 shows the examples of HISEA-1 and Gaofen-3 SAR images, highlighting their ability to monitor ships at sea in complex backgrounds [5].

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Distinct imaging principles underlie optical remote sensing and SAR sensors, each offering specific advantages for ground observation. SAR sensors provide unparalleled all-weather detection capabilities, penetrating clouds and fog, while optical remote sensing excels at conveying intuitive information such as shape, color, and texture. However, SAR's texture and feature radiation clarity often fall short, while optical sensors face constraints in data acquisition due to sunlight and weather conditions. Nonetheless, optical images extract rich spectral information, providing a robust foundation for interpretation. Efficiently interpreting vast volumes of high-resolution remote sensing images and harnessing information from multisource data are pivotal in remote sensing applications.

Aerospace reconnaissance equipment currently integrates advanced capabilities, including high-resolution ground imaging and multisource functionalities. Nonetheless, certain aspects of intelligence applications, such as SAR image interpretation and emergency reconnaissance, require enhancement. Target detection technology plays a pivotal role in transforming reconnaissance image data into actionable intelligence. Furthermore, the rapid development of small-satellite constellations has led to increased utilization of spaceborne SAR imagery for sea observation and extensive research on SAR ship detection [single shot multibox detector (SSD)] [6], [7], [8], [9], [10], [11], [12], [13].

Limited ship target image samples may be available at the data efficiency level. Effectively utilizing these data for training and learning can enhance the model's generalization ability. In military or security domains, data privacy and confidentiality are crucial, and using few-shot samples can minimize the risk of data leakage. Rapid deployment of ship detection algorithms is critical for effectiveness, and few-shot learning can accelerate training and improve system deployment. Few-shot learning also enhances adaptability to changes in various environments and imaging conditions, benefiting military reconnaissance. Real-time ship detection is crucial for effective harbor surveillance, and few-shot learning's adaptability makes it a valuable tool. Therefore, conducting SAR image ship detection under few-shot conditions is significant.

Existing deep learning models require extensive training data for optimal performance. In SAR image ship detection scenarios, a single sensor's limited measurement accuracy and operating conditions cannot meet the requirements for uncertainty detection in complex environments. Therefore, combining information from multisource heterogeneous sensors is crucial to enhancing SAR image ship detection

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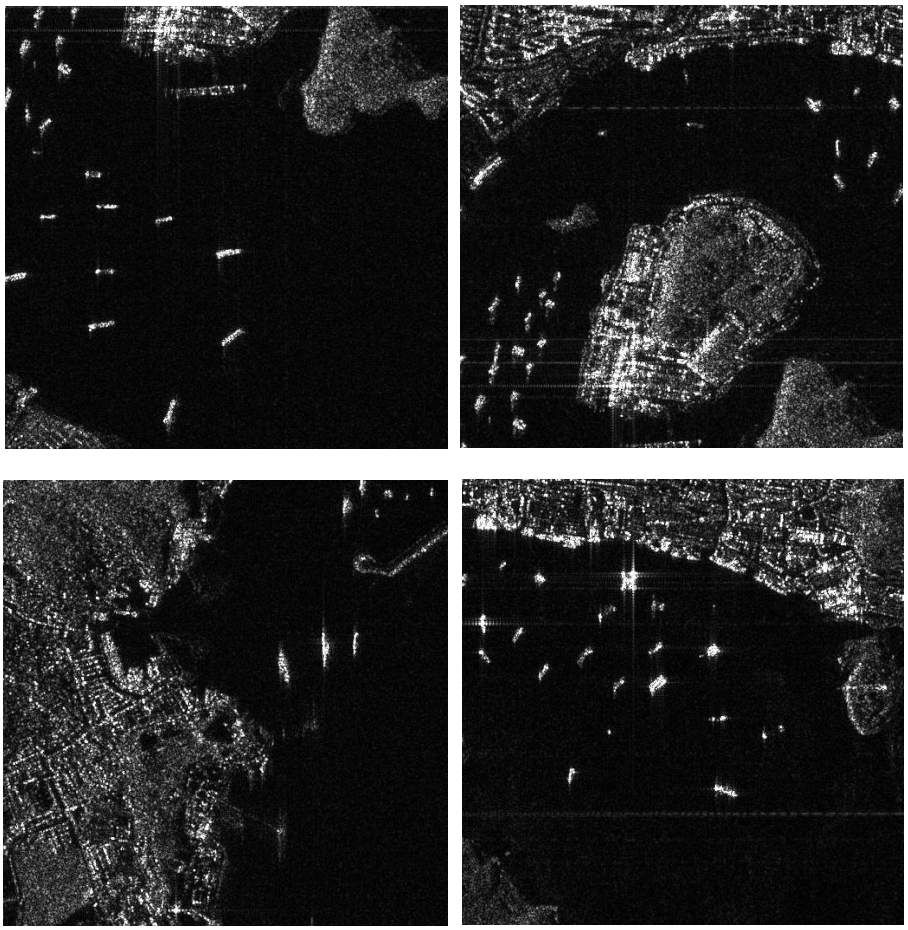


Fig. 1. Sample SAR ship images in the MSAR-1.0 dataset.

efficiency under few-shot conditions. Utilizing images from optical remote sensing sensors and SAR radar sensors for key feature information can effectively improve the few-shot SAR image ship detection capability through cross-domain implementation.

Due to domain differences, SAR images often degrade in performance when applied to detectors trained on optical images. Domain adaptation can facilitate generalization for detection by leveraging few-shot target detection across domains. Many existing unsupervised domain-adaptive target detection studies rely on feature alignment approaches, but they do not fully utilize ample data in the source domain. Some approaches focus on overcoming the domain gap in the unsupervised domain adaptation setting. However, certain techniques relying on simulated images fail to consider true distributional disparities between measured SAR images, making them unsuitable for practical applications. This article proposes a solution that directly combines limited SAR image data with sufficient optical image data, expanding it without tedious intermediate preprocessing.

The considerable disparities between optical and SAR domains have restricted the development of methods specifically tailored for investigating domain-adaptive techniques. Existing techniques are primarily focused on optical images due to shared optical domains. These differences primarily stem from variations in noise, weather conditions, and simulation factors. Addressing domain adaptation from optical to SAR images requires accurately measuring distributional differences and transferring domain knowledge. However, existing algorithms fall short in this regard. Accurately assessing distributional disparities between optical and SAR domains is crucial for guiding the network in acquiring domain knowledge. This article introduces OS-DAFSD, a domain-adaptive few-shot SSD algorithm, driven by potential similarities between optical and SAR images. It aims to accurately measure disparities across domains, guiding target images in learning domain knowledge from source images and ultimately enhancing target detection accuracy. The primary contributions of our work can be summarized as follows.

1) A domain-adaptive few-shot SAR ship detection algorithm was proposed, called OS-DAFSD, driven by the latent similarity between optical and SAR images. It achieves this by selecting optical ship images similar to SAR images to augment the data. It overcomes the constraint of conventional multisource information utilization that necessitates paired data and achieves cross-domain learning for unpaired data.
2) To tackle the challenge of transferring knowledge from optical images to SAR images, we propose a new distance metric function called optical domain to SAR domain (OSD) distance for domain-adaptive ship detection. OSD provides a more accurate distance metric

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between optical and SAR domains by constructing an efficient plug-and-play function.

3) To effectively utilize the similarity between the two domains, we designed a lossy branching domain knowledge loss (DKL) inspired by the latent similarity in domain knowledge distribution. These metrics measure the latent similarity between optical and SAR images, facilitating more effective learning of domain knowledge by the network.
4) To enhance the utilization of domain knowledge from SAR images under few-shot conditions, we constructed a weight-sharing, dual-stream, branching feature alignment extraction network called DFANet. This network facilitates better knowledge extraction and sharing between the two domains.
5) We constructed an adaptive ship detection dataset (DIOR2SSDD) extracted from a publicly available source. Our proposed method has been extensively validated through experiments, confirming its effectiveness.

The remaining sections are organized as follows. Section II provides a brief overview of recent relevant work. Section III elaborates on the proposed methods in detail. In Section IV, we introduce the reference datasets, experimental setup, and results. Section V discusses what knowledge to transfer, its limitations, and future work. Finally, Section VI concludes the thesis.

II. Related Work

In this section, we provide a concise review of research on SSD using both traditional methods and deep learning methods, along with an introduction to various domain adaptation methods. Finally, we summarize the limitations of current domain adaptation methods and propose the concept of addressing few-shot SSD through domain adaptation in this article.

A. Traditional Methods

Before the widespread adoption of deep learning in SAR target detection, researchers primarily used traditional methods. These methods laid a strong foundation for SSD advancements. Traditional SSD and recognition involve three phases: detection, identification, and recognition [14]. In the detection phase, traditional techniques extract noteworthy targets while mitigating false alarms from SAR images. The constant false alarm rate (CFAR) algorithm [15] is a prominent method, relying on clutter statistics and threshold values to identify SAR image features. Enhancements to CFAR include cell-averaging CFAR (CA-CFAR) [15], [16], and log normal distribution-based CFAR (LN-CFAR) [17], which improve false alarm detection, enhancing target identification and the practicality of SAR image recognition. During the identification stage, domain experts distinguish between true targets and false alarms [18], [19], [20] using their experience and knowledge, considering geometric details, semantic characteristics, and physical insights. The final recognition stage involves model-based and template-matching-based approaches [21]. In the model-based approach, an image generation model is created through electromagnetic model simulation. This model generates highly accurate simulated images that are iteratively matched with the test image to achieve recognition [22]. Conversely, the template-matching-based approach extracts salient features from the SAR image and uses classifiers such as similarity measures, support vector machines, hidden Markov models, nearest neighbors, Bayesian models, and neural networks from template libraries to achieve automatic target type recognition [21].

B. Deep Learning Methods

Traditional methods are unsuitable for complex SAR image backgrounds or significant changes in target scale, often relying on manual parameter selection. However, deep learning technology has vastly improved SAR ship target detection. Experts and scholars worldwide have achieved significant strides in this field [2], [3], [4], [7], [8], [23], [24], [25] focusing on continuous innovation to meet diverse requirements. Moreover, the release of numerous SAR image ship datasets in recent years has laid a robust groundwork for SAR image target detection and recognition [5], [26], [27], [28].

Deep learning methods for SAR image target detection are broadly categorized into anchor-based and anchor-free types.

Anchor-based methods are further divided into single- and two-stage approaches. Single-stage methods directly detect targets within input images, predicting target coordinates and category information for accurate detection. Examples include SSD, feature pyramid network (FPN), and you only look once (YOLO) [29]. In contrast, two-stage methods involve region proposal generation, feature extraction, and categorization prediction steps, with prominent examples being RetinaNet [30], Faster R-CNN [31], and Mask R-CNN [32].

Anchor-free methods establish spatial connectivity among key points or represent targets using a center point, with notable examples, including YOLOX, CenterNet, CornerNet, and fully convolutional one-stage (FCOS) [29]. Researchers have integrated SAR ship image characteristics with deep learning frameworks [2], [33], [34], [35], incorporating innovative designs such as attention mechanisms, multiscale feature fusion, and loss functions. These endeavors aim to refine and optimize deep learning methods for SAR image target detection, thereby enhancing their performance or applicability.

C. Domain Adaptation Methods

Acquiring SAR images poses unique challenges compared to obtaining optical images. Training an accurate SAR image target detector requires a substantial volume of SAR training data. Therefore, using optical images in the same scene to assist SAR target detection and recognition with unified category types is intuitive. Domain-adaptive target detection and recognition effectively address this issue [36]. It only necessitates matching target types in similar scenes with comparable shapes and sizes, without stringent resolution requirements, rather than fully registered or co-temporal optical and SAR image pairs. Domain-adaptive target detection effectively utilizes information from mismatched heterogeneous images. The change detection method for heterogeneous images that also

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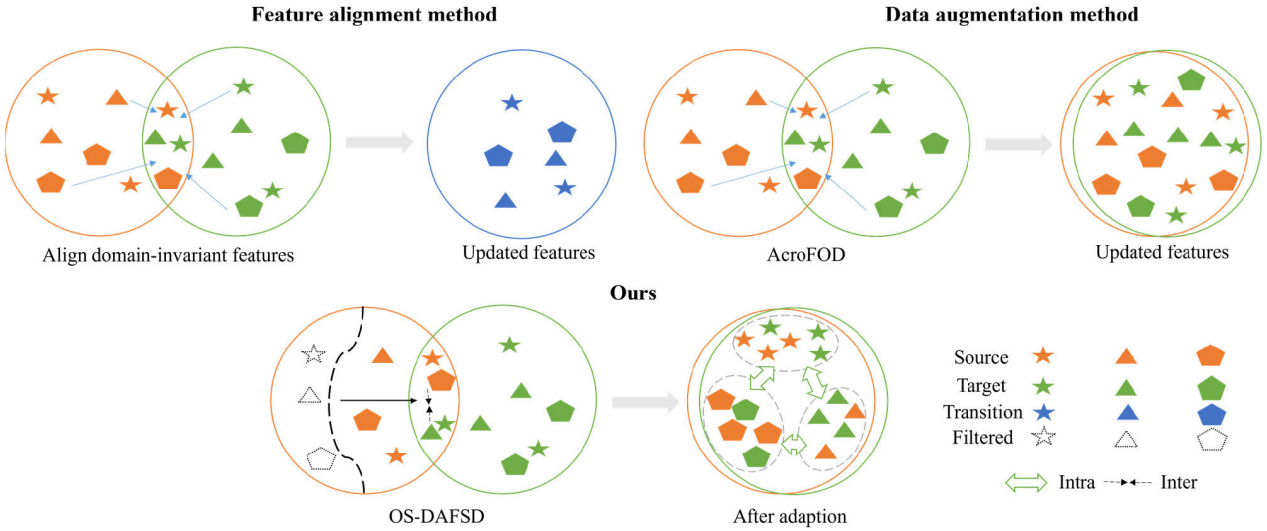


Fig. 2. This article discusses detecting ships in few-shot SAR images across different domains. Existing feature alignment-based methods are inadequate in extracting discriminative features from the limited labeled SAR data. Similarly, existing data augmentation-based methods are unable to learn the discriminative features from the two domains. Our method effectively combines the advantages of feature alignment and data enhancement methods to filter the source data with large distribution differences from the target domain (left of the thick black dashed line), which can learn the discriminative features from the two domains while aligning them.

utilize optical and SAR images requires fully registered image pairs [37]. Therefore, designing a good heterogeneous image matching [38] method is also essential, and our proposed method does not require complex preprocessing processes.

While numerous domain adaptation studies between optical images to SAR images have been conducted recently, the most common solution is featuring alignment. DAF [39] employed adversarial feature alignment methods [40], [41] for alignment at both the image and instance levels. Graph-induced prototype alignment (GPA) [42] investigated the effectiveness of graph-induced prototypes and class-weighted contrast loss for feature alignment within a two-stage training framework. Kim et al. [43] proposed a domain-adaptive target detector based on the single-stage SSD detector and investigated the effectiveness of single-stage self-training and adversarial background regularization. Gao et al. [44] introduced a few-shot domain-adaptive target detection method based on self-adaptive data filtering through iterative distribution optimization and multilevel domain-aware enhancement for cross-domain target detection tasks. However, these methods are experimental studies conducted on optical images, with few related studies on domain-adaptive target detection for SAR images. Chen et al. [45] and He et al. [46], [47] conducted studies on target recognition for SAR images using domain adaptation, focusing on classification and recognition tasks rather than detection, and the domain adaptation process was from SAR image to SAR image. Similarly, Guo et al. [48], Liao et al. [49], and Shi et al. [50] explored optical to SAR domain adaptation in SAR image target detection, primarily using Faster R-CNN [31] for feature alignment. Pu et al. [51] proposed an unsupervised domain adaption SSD method.

All the mentioned methods aim to deal with the challenge of adapting different types of images, but aligning optical images with SAR images faces a big hurdle due to their different distributions. Overcoming this involves narrowing down or measuring the gap between their distributions. Also, current methods overlook the fact that SAR images often have limited examples and do not fully use the information in optical images to help in SAR image target detection with few examples. To fix this, one could train the detector using images from both the two domains. However, combining data in this way risks adding a lot of poor-quality, irrelevant data to the target domain, which could harm the model's training. Thus, this article introduces a new algorithm, OS-DAFSD. It filters out harmful source domain images and combines information from both domains. It also includes feature alignment techniques to enhance the model's generalization capability and address issues with inconsistent environmental data distribution. As shown in Fig. 2, OS-DAFSD facilitates rapid and accurate adaptation to the feature distribution of the target domain, leading to better performance in few-shot SAR image ship detection.

III. Methodology

This section proposes a domain-adaptive few-shot SAR ship detection algorithm driven by the latent similarity between optical and SAR images proposed in this article. This section provides a detailed introduction to the functions and design ideas of each module proposed in this article.

A. Overview of OS-DAFSD

Fig. 3 illustrates the network framework of the OS-DAFSD algorithm. This study begins with a focus on data enhancement, crafting an adaptation framework for multisource information domains. It integrates texture and structured details from optical images as cross-domain prior knowledge, directing the cross-domain learning model toward salient characteristics of optical image targets. In addition, it builds an

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SSD model via the feature alignment method, incorporating data-domain-aware adaptive enhancement and weight sharing, ultimately enhancing ship detection efficacy.

First, the article introduces the cross-domain few-shot target detection benchmark framework, AcroFOD [44], adapted into a model suitable for SAR images. Leveraging a vast dataset of labeled optical image ship targets, it aids ship detection in SAR images under few-shot conditions. Second, a data domain-aware self-adaptive enhancement strategy is devised to generate diverse candidate samples, addressing the challenge of insufficient and identical domain-aware enhancement samples. This strategy also mitigates distributional differences and domain bias. Subsequently, a novel distance metric function, called OSD, is designed to measure distributional differences between domains more effectively, enabling iterative and adaptive optimization of candidate image distribution in the target domain, thereby enhancing detector generalization. A loss branch DKL, guided by latent similarities in domain knowledge, optimizes the utilization of SAR domain knowledge. Finally, a weight-sharing dual-stream branching feature alignment extraction network, called DFANet, facilitates knowledge extraction and sharing between domains.

To effectively mitigate the distribution differences between optical and SAR images, the method in this section designs a domain-aware adaptive data enhancement strategy. The few-shot class SAR ship target data are fused and augmented with the optical ship target data with sufficient data volume. The source domain is assumed to be a dataset $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ with sufficient target volume, and the target domain contains only a few samples $D_t = \{(x_j^t, y_j^t)\}_{j=1}^{n_t}$, where $x_i^s, x_j^t \in \chi$ is the input image, $y_i^s, y_j^t \in Y$ consists of the target anchor box coordinates and target classes of x_i^s and x_j^t, and there is usually a discrepancy between the source domain D_s distribution $\mathrm{P}_s : \chi \times Y \to \mathrm{R}^+$ and the target domain D_t distribution $\mathrm{P}_t : \chi \times Y \to \mathrm{R}^+$ due to factors such as different imaging mechanisms.

To maximize the mining and utilization of the annotated target valid information in the source domain and to extend $D_t \sim \mathrm{P}_t$ and $D_s \sim \mathrm{P}_s$ to $\tilde{D}_t$, this section designs the domain-aware data enhancement strategy to generate the new data $\tilde{D}_t = Aug(D_s, D_t)$, to perform the multistrategy domain-aware adaptive data enhancement of the source data at the image level and the box level to approximate the distributions of the source and target domain data. Fig. 4 displays the schematic of domain-aware adaptive data enhancement at both the image and box levels. The proposed method involves randomly extracting a corresponding number of images from each of the two domains and then blending them into a single image for image-level domain-aware data enhancement. For box-level domain-aware data enhancement, ship instance labels are separated from the original SAR image background and placed on other regions of the optical image to efficiently utilize the limited number of SAR images available.

Assuming that the detector $f_\theta = (g_\theta, h_\theta)$ is defined by a set of parameters θ consisting of the backbone g_θ and the detector head h_θ, the representations of x_j^t in the output D_t and x_i^{aug} in $\tilde{D}_t$ are shown as follows:

$$z_i^{\mathrm{aug}} = g_\theta\left(x_i^{\mathrm{aug}}\right), z_j^t = g_\theta\left(x_j^t\right). \tag{1}$$

Furthermore, a metric function dist_f is defined to measure the distance between x_i^{aug} and $\{x_j^t\}_{j=1}^{n_t}$ to rank the enhancement candidate set x_i^{aug}

$$d_i^{\mathrm{aug}} = \mathrm{dist}_f\left(z_i^{\mathrm{aug}}, \left\{z_j^t\right\}_{j=1}^{n_t}\right). \tag{2}$$

The benchmark method in this article uses a maximum mean difference (MMD) [52] as the distance metric function

$$\mathrm{MMD}^2\left(x, x'\right) = \left\| \frac{1}{n_t} \sum_{j=1}^{n_t} x - x' \right\|_2^2. \tag{3}$$

Therefore, the distance measurement function of our baseline is ultimately defined as

$$\mathrm{dist}_f\left(x, x'\right) = \mathrm{MMD}^2\left(x, x'\right) = \left\| \frac{1}{n_t} \sum_{j=1}^{n_t} x - x' \right\|_2^2. \tag{4}$$

To filter the noisy samples in $\tilde{D}_t$, the original benchmark designs the shrinkage rate $k(0 < k \leq 1)$ to dynamically reduce the number of noisy samples, and furthermore, based on d_i^{aug}, a new optimization function ϕ_{opt} is constructed to optimize $\tilde{D}_t$ to obtain the optimization expansion domain $\tilde{D}_t{}^{\mathrm{opt}}$ as defined in the following:

$$\tilde{D}_t^{\mathrm{opt}} = \left\{\left(x_i^{\mathrm{opt}}, y_i^{\mathrm{opt}}\right)\right\}_{i=1}^{n_b} = \phi_{\mathrm{opt}}\left(\tilde{D}_t, \left\{d_i^{\mathrm{aug}}\right\}_{i=1}^{n_a}, k\right). \tag{5}$$

By ϕ_{opt}, $n_b (n_b = [n_a * k])$ selects candidates $\tilde{D}_t{}^{\mathrm{opt}}$ from $\tilde{D}_t$ in increasing order of d_i^{aug} to better reflect the target domain distribution P_t.

Simply put, the data distribution of the two domains is calculated to filter out source images that are dissimilar to the target images. This ensures that only source images with a similar distribution to the target images are used for training the network. Data that are dissimilar to the target domain are discarded as it can negatively impact the model's training.

B. Optical Domain to SAR Domain

The importance of a distance metric function in the domain adaptation landscape is paramount. This tool plays a crucial role in quantifying the distributional gap between two domains, functioning at either the data or feature level. A more effective and accurate distance measurement function can facilitate the transfer of domain knowledge from the source images to the target images. To tackle this challenge, this article sets out to design a distributional metric function tailored to the specific requirements of domain adaptation from optical images to SAR images. Importantly, after extracting features from the model, both optical and SAR images reside in a high-dimensional space that includes spatial angle information alongside magnitude distance information. This information signifies the distributional disparity between the two. Consequently, in the high-level feature space, it is imperative to incorporate the distributional metric of spatial angle information for both optical and SAR images. Nevertheless, the commonly used distance metric functions, such as the

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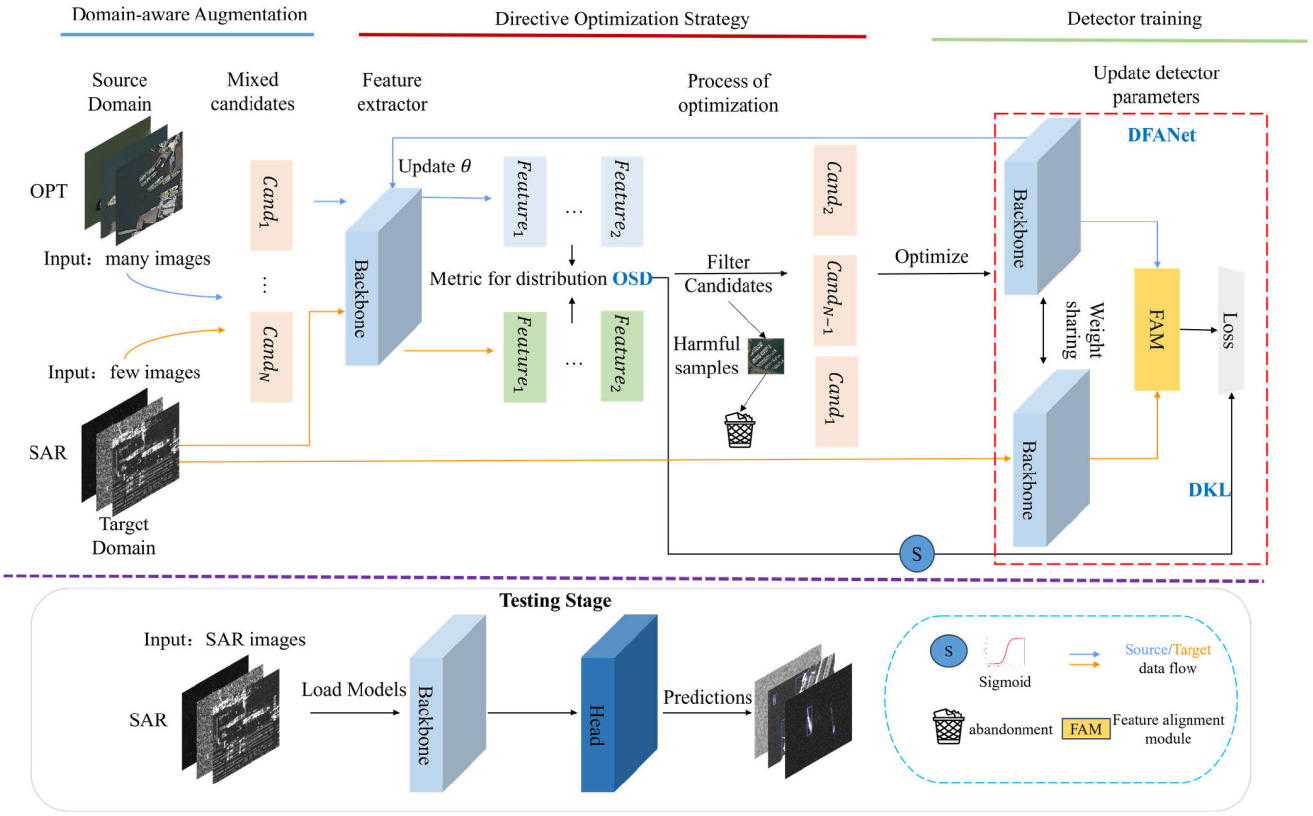


Fig. 3. Overall framework of the OS-DAFSD model.

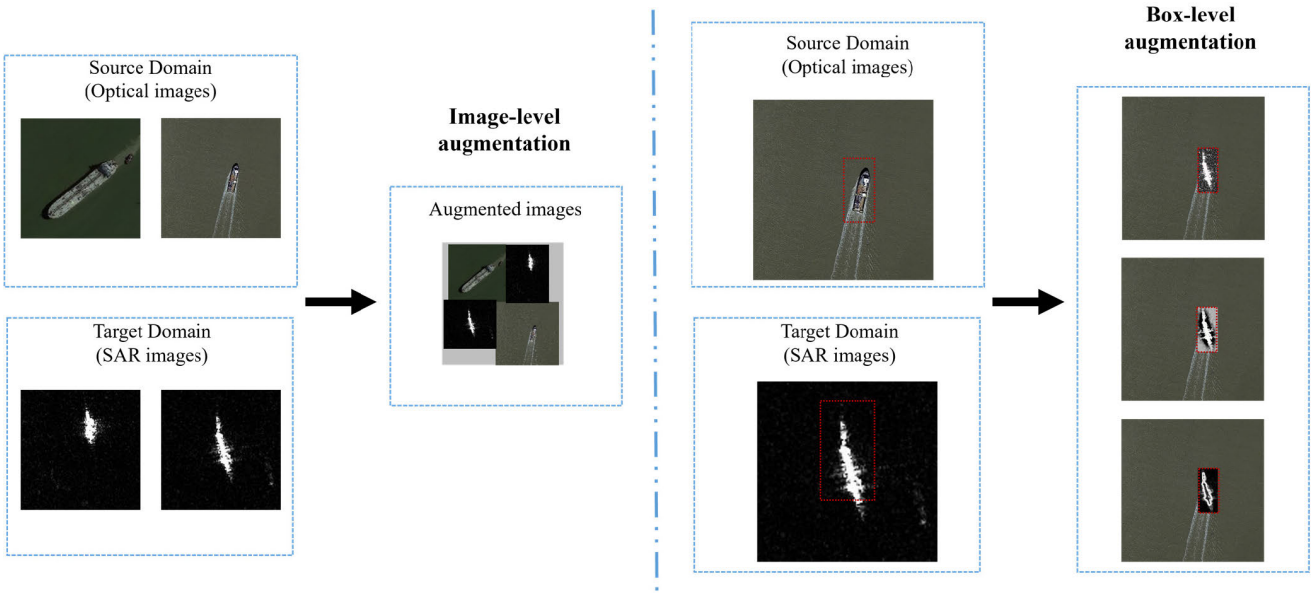


Fig. 4. Schematic of domain-aware adaptive data augmentation at the image and anchor box levels.

Euclidean distance, MMD [43], and cosine distance (CSD) [44], [45], [46], are limited in their ability to assess specific aspects of this intricate relationship. The Euclidean and MMD metrics assess disparities in magnitude distribution, whereas the CSD metric concentrates on differences in the distribution of direction and angle within the high-dimensional features of the two domains. However, it fails to measure the difference in the distribution of magnitude and distance between the two domains. However, neither fully captures the complete extent of the distributional divergence between the two domains. Therefore, the central research question of this article is focused on effectively integrating these two critical aspects. Drawing inspiration from the principles of complex functions and signal systems, this article explores the common

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formulas that govern complex numbers. This approach lays the groundwork for novel solutions that incorporate both amplitude and spatial direction dimensions, thus offering a more comprehensive distance metric function capable of tackling the intricate challenges of domain adaptation. The common formula for complex numbers is

$$Z = A + Bi. \tag{6}$$

Its exponential function takes the form

$$Z = \mathrm{R}e^{i\theta} \tag{7}$$

where R is the modulus of Z, θ is the principal value of the angle, $A = \mathrm{R}\cos\theta$, $B = \mathrm{R}\sin\theta$, and $\mathrm{R} = (A^2 + B^2)^{1/2}$.

From the exponential form of Euler, R is the magnitude information, so we can represent it as the magnitude distribution distance of the domain adaptation metric function, while θ is the angle information, which we can represent as the angle direction distribution distance between high-dimensional features. Thus, we can take MMD as the magnitude part R and CSD as the angle part θ. The CSD metric functions are shown as follows:

$$\mathrm{CSD}(x, x') = \cos(x, x') \tag{8}$$

where x and x' indicate two different distributions. This defines our proposed distribution distance metric function, denoted $\mathrm{dist}_{\mathrm{OSD}}$. From the above derivation and definition, the $\mathrm{dist}_{\mathrm{OSD}}$ formula is instantiated as follows:

$$\mathrm{dist}_{\mathrm{OSD}}(x, x') = \left\| \frac{1}{n_t} \sum_{j=1}^{n_t} x - x' \right\|_2^2 e^{\cos(x, x')}. \tag{9}$$

Therefore, we have developed an efficient distance metric function called OSD distance for domain-adaptive ship target detection. OSD is better suited for measuring the distribution difference between the source domain and the target domain. To improve the accuracy of ship target detection in SAR images through multisource image information utilization, we use OSD as the distribution metric for the algorithm. OSD replaces $\mathrm{dist}_f(x, x')$, which was described in Section III-A. OSD not only considers the distances between domains in the high-dimensional feature space but also takes into account the direction of the distributions between domains. This better aligns the relationships and knowledge between different data domains.

Fig. 5 shows the amplitude distance measurement between points A and B in high-level semantic features, as well as the directional angle measurement.

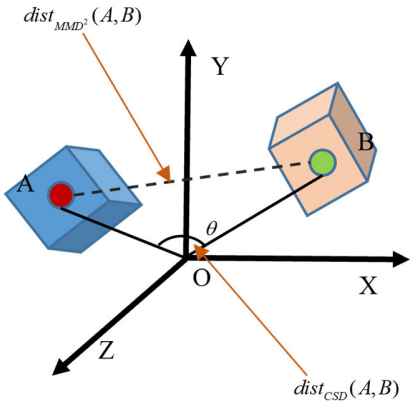


Fig. 5. Schematic of distance measurement between two points in high-level semantic features.

C. Domain Knowledge Loss

According to the algorithm framework depicted in Fig. 3, it is apparent that the algorithm employs the metric function dist_f to assess the distribution of the two domains via the backbone. Subsequently, the algorithm filters out the source images that are closer in distribution to the target images. However, this strategy has limitations as it overlooks the potential impact of differences in data distribution within each domain batch. To better leverage the disparities between optical and SAR images, we devised a loss branch, DKL, guided by the latent similarities in the distribution of domain knowledge. This branch employs a metric to quantify the distributional disparity, thereby facilitating more effective learning of domain knowledge by the network. The equation for the loss function is

$$\mathcal{L}_{\mathrm{det}} = \mathcal{L}_{\mathrm{cls}} + \mathcal{L}_{\mathrm{obj}} + \mathcal{L}_{\mathrm{box}} \tag{10}$$

where $\mathcal{L}_{\mathrm{cls}}$ is the class loss function of the model, $\mathcal{L}_{\mathrm{obj}}$ is the confidence loss function of the target object, and $\mathcal{L}_{\mathrm{box}}$ is the bounding-box loss function. Since only ship targets are detected in this article, the class loss is 0 here, which is obtained

$$\mathcal{L}_{\mathrm{det}} = \mathcal{L}_{\mathrm{obj}} + \mathcal{L}_{\mathrm{box}}. \tag{11}$$

From Section III-A, we define the metric function dist_f to measure the distance d_i^{aug} between each batch x_i^{aug} and $\{x_j^t\}_{j=1}^{n_t}$, which is averaged and scaled by the sigmoid function between 0 and 1 to obtain $\mathcal{L}_{\mathrm{dkl}}$

$$\mathcal{L}_{\mathrm{dkl}} = \mathrm{sigmoid}\left(\mathrm{mean}\left(d_i^{\mathrm{aug}}\right)\right) \tag{12}$$

where $\mathrm{sigmoid}(x) = (1/1 + e^{-x})$, which is the excitation function in CNN, is a smooth and strictly monotonic saturation function and easy to derive, often used in the hidden layer neuron output, the range of values for (0, 1), and its function curve shown in Fig. 3, due to the algorithm in d_i^{aug} that is greater than 0 values, so the value of $\mathcal{L}_{\mathrm{dkl}}$ range is (0.5, 1). We add $\mathcal{L}_{\mathrm{dkl}}$ to the final loss for backpropagation

$$\mathcal{L}_{\mathrm{det}} = \mathcal{L}_{\mathrm{obj}} + \mathcal{L}_{\mathrm{box}} + \mathcal{L}_{\mathrm{dkl}}. \tag{13}$$

DKL utilizes domain knowledge distribution latent similarity to drive and inspire the training of network models, leading to better learning of domain knowledge.

D. DFANet

Due to the abundance of data for optical images, there may be a temptation to utilize the entire dataset from both domains to train the detector. The previously discussed latent similarity in the distribution of domain knowledge motivated us to filter potential strategies for selecting source domain optical images. This approach effectively mitigates domain mismatch

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between optical and SAR images, removes detrimental source domain images, and enhances the diversity and rationality of knowledge fusion. However, this statement solely focuses on enhancing data through the use of optical data and fails to address the disparities in features extracted from optical and SAR images. The full potential of SAR images' unique characteristics may not be realized if training relies solely on filtered source data. To facilitate domain knowledge transfer from optical images in the source domain to shot-less SAR images in the target domain, we developed DFANet, a weight-sharing dual-stream branching feature alignment extraction network. This network enables knowledge extraction and sharing across both domains. This approach enables the utilization of a small number of instances in the target domain to guide the extraction and transfer of domain knowledge from a larger number of instances in the source domain. The specific process can refer to Fig. 3, and during network training, we combined a limited amount of SAR data from the target domain with an ample supply of optical data from the source domain. In actual model testing, only SAR images need to be used as input. After training, only test the model's performance on SAR images and evaluate the results. In addition, we input the limited SAR data from the target domain into another feature extraction backbone that shares the same features as the source domain for feature extraction. The feature extraction backbones of both domains share weights, enabling mutual learning and feature alignment. Subsequently, the corresponding loss is generated for backpropagation and parameter updating, ultimately leading to domain adaptation, as illustrated in Fig. 2.

As described in the previous sections, the total loss function for target detection is already known. In addition, we have constructed a weight-sharing instance-level feature extraction branch backbone for feature extraction and alignment of few-shot SAR images in the target domain. This increases the confidence and bounding-box loss on the SAR image in the target domain, resulting in the current total loss function for target detection

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{obj}} + \mathcal{L}_{\text{obj}}^{\text{tgt}} + \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{box}}^{\text{tgt}} + \mathcal{L}_{\text{dkl}} \tag{14}$$

where $\mathcal{L}_{\text{obj}}^{\text{tgt}}$ denotes the confidence loss function of the SAR image ship target on the target domain and $\mathcal{L}_{\text{box}}^{\text{tgt}}$ denotes the bounding-box loss function of the SAR image ship target on the target domain.

Furthermore, assign certain ground weights to the confidence and bounding-box losses in the two domains to construct the total loss function as follows:

$$\min_{F_\theta} \mathcal{L}_{\text{det}} = \min_{F_\theta} \lambda_1 \left(\mathcal{L}_{\text{obj}} + \mathcal{L}_{\text{obj}}^{\text{tgt}} \right) + \lambda_2 \left(\mathcal{L}_{\text{box}} + \mathcal{L}_{\text{box}}^{\text{tgt}} \right) + \mathcal{L}_{\text{dkl}} \tag{15}$$

where F_θ denotes the parameterized model and λ_1 and λ_2 are the weighting parameters between the target detection loss function and the domain adaptation loss function, which are set to 1 in the later experiments.

IV. Experiments

This section aimed to validate the effectiveness of a domain-adaptive few-shot SAR ship detection algorithm, OS-DAFSD, driven by the latent similarity between optical and SAR images proposed in the article. To achieve this, we constructed the DIOR2SSDD cross-domain dataset by combining the public benchmark SSD dataset (SSDD) [26], with the large multiclass target optical remote sensing dataset DIOR [53]. We conducted numerous experiments on this dataset to validate our approach. In addition, we compared our proposed method, OS-DAFSD, with existing common detection models, as well as excellent domain-adaptive detection models, to demonstrate its superior performance.

A. Dataset Introduction and Processing

We selected the common remote sensing image dataset DIOR [53] and SAR image datasets SSDD [26] as the two domains for constructing cross-domain datasets and further processed them.

1) The DIOR dataset, a creation of Google, stands as a substantial benchmark dataset meticulously crafted for target detection within optical images. Sourced from Google Earth, this dataset offers an expansive compilation of visual data to facilitate comprehensive research and development endeavors. Across its impressive expanse, the DIOR dataset encompasses a total of 23 463 distinct images. Within these images, a total of 192 472 instances belonging to 20 distinct categories are meticulously annotated. This comprehensive labeling ensures that the dataset spans an array of target types, enhancing its versatility and applicability. The 20 distinct target categories represent a broad spectrum of objects, ranging from ships to bridges and oil tanks, among others. Each image within the DIOR dataset maintains an image size of 800 × 800, while the spatial resolution exhibits a dynamic range spanning from 0.5 to 30 m. Notably, the dataset's defining characteristic is the marked variability in target sizes, contributing to its realism and practical relevance. Given its extensive scope, the DIOR dataset offers an invaluable resource for researchers and practitioners involved in the domain of target detection within optical images. The wealth of diverse instances, coupled with meticulous annotations, positions the DIOR dataset as an indispensable tool for pushing the boundaries of knowledge and innovation in this field.
2) The SSDD dataset is a publicly accessible repository specifically curated for small-ship target detection in SAR images. This dataset encompasses a diverse array of target sizes, with a predominant focus on smaller targets. Within the SSDD dataset, SAR images were captured by satellite sensors, encompassing four distinct polarizations: HH, HV, VH, and VV. These images boast impressive resolutions ranging from 1 to 5 m. Comprising a total of 1160 images, the SSDD dataset offers a rich array of samples for analysis. On average, each image contains approximately 2.12 ships. The dataset is thoughtfully partitioned in a ratio of approximately 8:2, with 928 images earmarked for training purposes and the remaining 232 images allocated for testing. An essential attribute of the SSDD dataset is

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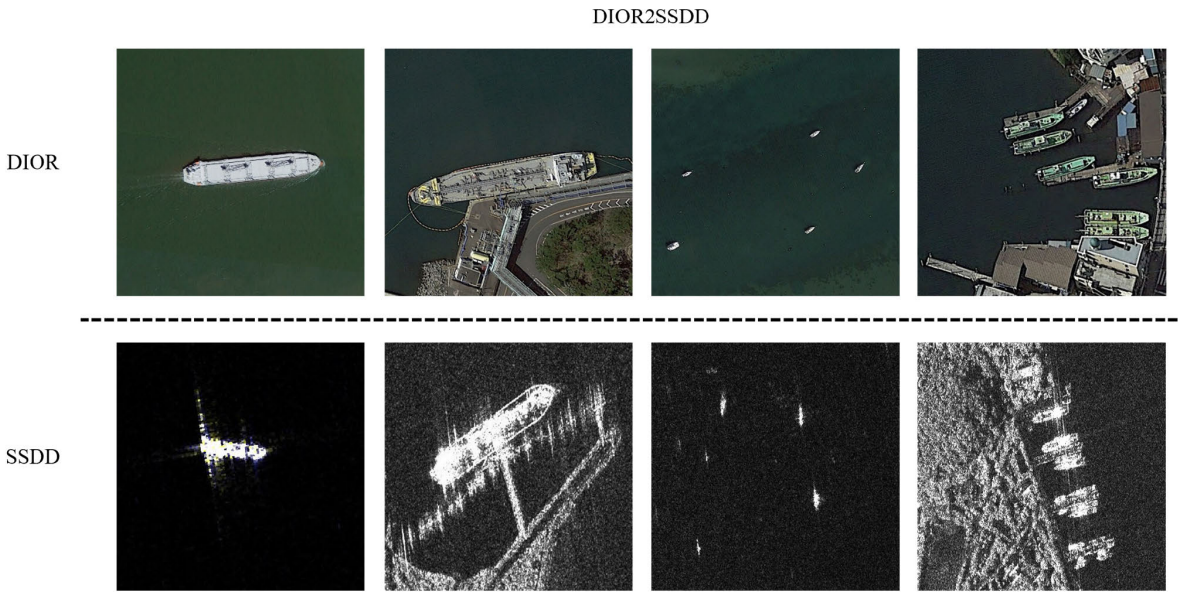


Fig. 6. Some examples from the self-built optical image to SAR image domain adaptation dataset DIOR2SSDD.

TABLE I
DETAILS OF THE TWO SELF-CONSTRUCTED OPTICAL IMAGES TO SAR IMAGES' DOMAIN ADAPTATION DATASETS

	train: valid	number of class	class	Number of source training	Number of target training	Number of target valid
DIOR2SSDD	8:2	1	ship	2161	928	232

its alignment with existing data distributions, fostering equitable, and impartial comparisons of results. The SSDD dataset stands as a valuable resource for researchers and practitioners engrossed in the realm of small-ship target detection using SAR imagery. Its comprehensive coverage of target sizes, polarization modes, and resolutions, coupled with a balanced division for training and testing, ensures its utility in benchmarking and advancing the field of SAR image target detection techniques.

1) DIOR2SSDD (Optical → SAR): We preprocessed the remote sensing dataset DIOR since it contains multiple classes of targets, including ships, unlike the SSDD dataset that only has one class of ship targets. All the images containing ship targets are extracted from the DIOR dataset and divided into a training set and a validation set in an 8:2 ratio. The training set comprises 2161 optical images. The source images consist of all images containing ships taken from the optical remote sensing dataset DIOR, while the target domain comprises SAR images containing ships from the training set of SSDD. The number of SAR images to be extracted as few-shot samples (three-, five-, and ten-shot settings) is determined based on the number of targets involved in the designed model for experiments related to domain adaptation-based target detection.

Fig. 6 presents a diagram of the two self-constructed datasets that adapt optical images to SAR images. Table I shows the details specific to the datasets.

B. Experimental Setup

The proposed OS-DAFSD algorithm, a domain-adaptive few-shot SAR ship detection algorithm driven by the latent similarity between optical and SAR images, was validated through comprehensive experiments. Our model is compared with the original baseline model, classical target detection methods, and related domain-adaptive algorithms. For the experiments on classical target detection algorithms, source and target data are combined and fed directly into the model for training. Our OS-DAFSD has similar experimental parameter settings to the benchmark. Both models are trained on the cross-domain few-shot datasets described earlier and then validated on the same target domain validation set. To compare with other classical target detection models, the training dataset is selected to include the desired number of targets. Subsequently, optical images and few-shot SAR images are combined and utilized to train the classical target detection model. This validates the effectiveness of our domain-adaptative algorithm.

To ensure that the performance of OS-DAFSD is validated under fair experimental conditions, the training parameters of the OS-DAFSD model were kept the same as those of the baseline model and YOLOv5. The YOLOv5n was selected for the model profile with version 3.0. The batch size was 16, the size of the trained image was 640 × 640, and the number of training epochs was 300, while the number of targets in the target domain training images is set to three-, five-, and ten-shot, as described in Section IV-A. The remaining comparison experiments are kept consistent. The graphics card is NVIDIA GeForce RTX 2060 Super, and the experimental environment is Windows 10, CUDA 11.1, PyTorch 1.7.1, and Python 3.7. The remaining comparative experiments aim to maintain parameter consistency by maintaining default values or as consistent as possible.

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C. Evaluation Index

We rigorously evaluate the effectiveness of our proposed method using the widely accepted VOC [54] evaluation criteria. Specifically, we employ the mean average precision (mAP) metric at a threshold of 0.5, denoted as mAP50, as the primary evaluation metric. This metric serves as a pivotal yardstick to gauge the effectiveness of our approach in both experimental benchmarks. The calculation of mAP50 entails a thorough assessment of accuracy P and recall R. This comprehensive evaluation accounts for the true positive (TP), false positive (FP), and false negative (FN) instances within the detection process. By harmonizing precision and recall, mAP50 delivers a holistic measure of our method's detection performance, providing insights into its accuracy, sensitivity, and overall effectiveness. By adopting the mAP50 metric at a threshold of 0.5, we ensure a consistent and standardized approach to evaluating the performance of OS-DAFSD across the experimental benchmarks. This meticulous evaluation framework enables an impartial and informative comparison of our approach against existing methods, underscoring its potential to enhance target detection accuracy and applicability.

Certainly, the accuracy rate is a fundamental evaluation metric that measures the proportion of correctly predicted positive cases within a given dataset. This metric provides insight into the model's capability to accurately identify instances of interest. The accuracy rate is calculated as the ratio of TP predictions (correctly identified positive cases) to the total number of samples predicted as positive, which includes both TPs and FPs

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{16}$$

The recall, however, indicates the ratio of predicted positive cases to actual positive cases, and it is calculated by dividing TPs and FNs

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{17}$$

The mAP is calculated as the integral area enclosed by the precision–recall (PR) curve

$$\mathrm{mAP} = \int_0^1 \mathrm{P}(\mathrm{R}) d\mathrm{R}. \tag{18}$$

D. Experimental Results

We conducted numerous experimental analyses on the self-constructed domain-adaptive dataset of optical images to SAR images, DIOR2SSDD, to assess the effectiveness of OS-DAFSD, a domain-adaptive few-shot SAR ship detection algorithm driven by the latent similarity between optical and SAR images proposed in this article. From a quantitative analysis perspective, we compared the experimental results of OS-DAFSD with various excellent domain-adaptive methods [39], [55], [56], including baseline model [44], and classical target detection methods [32], [57], [58]. In addition, a lot of ablation experiments on its dataset were conducted to validate the effectiveness of the proposed individual modules. Second, in terms of qualitative analysis, we visualize the detection results and compare them with other models.

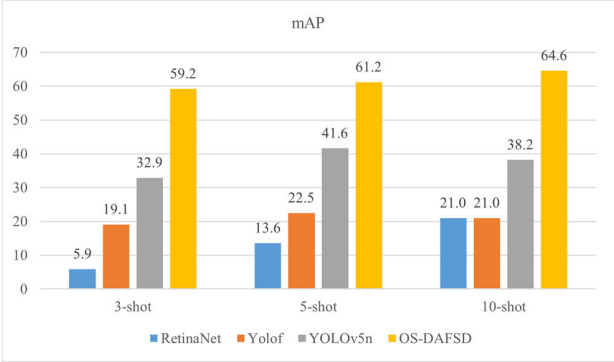


Fig. 7. Graph of Table II.

We visualized the feature distributions of the two domains using t-SNE [59] to verify whether the distributions of the screened optical images are close to those of the few-shot SAR images. In addition, we analyzed the validity of the OS-DAFSD by visualizing the feature heatmaps using the grad-cam [60].

1) Comparison Results of OS-DAFSD on DIOR2SSDD: Table II presents the experimental results of OS-DAFSD, a few-shot domain-adaptive SSD algorithm, on the DIOR2SSDD dataset. The results validate and compare OS-DAFSD with several classical target detection models on the DIOR2SSDD dataset in numerous experiments. Our method demonstrates excellent few-shot ship detection performance during the domain adaptation. Fig. 7 displays the bar charts of the results in Table II.

Table II displays the effectiveness of OS-DAFSD, a domain-adaptive few-shot SAR ship detection algorithm driven by the latent similarity between optical and SAR images, as designed in this article. OS-DAFSD's performance in domain-adaptive few-shot ship detection from optical to SAR images significantly surpasses that of classical target detection models listed in the table, with mAP improving by over 20 points compared to YOLOv5n under each shot setting. The limitations of directly mixing optical image data and SAR image data for training are evident. This verifies that the data enhancement method is ineffective, and arbitrary data enhancement may even impact the original target domains and increase the differences between the domains. Harmful samples can damage the distribution of the target data. Thus, the experimental results indicate that the superiority of the OS-DAFSD we proposed.

In addition to experimental validation comparisons on classical target detection methods, we have also conducted extensive experiments on a variety of excellent domain-adaptive methods, including the baseline model, and compared the experimental results with the OS-DAFSD. Tables III–V show the comparison results under three different shot settings.

As the results indicated in Tables III–V, the OS-DAFSD is not only optimal in P and R metrics under three different few-shot settings but also substantially outperforms some excellent domain-adaptation methods in mAP metrics and compares with the suboptimal AcroFOD algorithm [44]; OS-DAFSD improves at least 5% points in mAP and up to 8.7% points in the three-shot setting. This demonstrates the effectiveness as well as the superiority of the OS-DAFSD

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TABLE II
COMPARISON RESULTS WITH CLASSICAL TARGET DETECTION METHODS

Methods	ship AP50 (%)		
	3-shot	5-shot	10-shot
RetinaNet	5.9	13.6	21.0
Yolof	19.1	22.5	21.0
YOLOv5n	32.9	41.6	38.2
OS-DAFSD	**59.2**	**61.2**	**64.6**

TABLE III
COMPARISON RESULTS WITH EXCELLENT DOMAIN-ADAPTIVE TARGET DETECTION METHODS UNDER THREE-SHOT SETTING

Methods	3-shot ship (%)		
	P	R	mAP
DA	31.5	24.2	16.5
HTCN	27.2	29.3	17.1
SSDA-YOLO	31.5	27.8	19.2
AcroFOD	62.6	49.3	50.5
OS-DAFSD	**72.4**	**56.0**	**59.2**

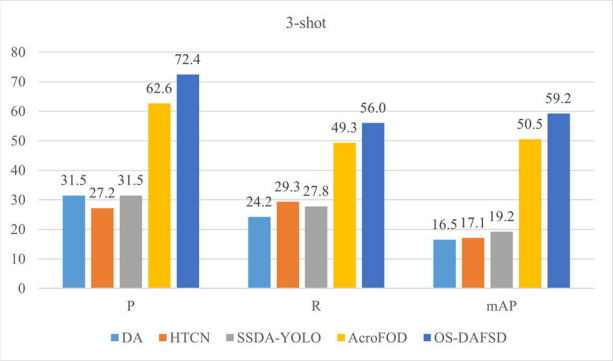


Fig. 8. Graph of Table III.

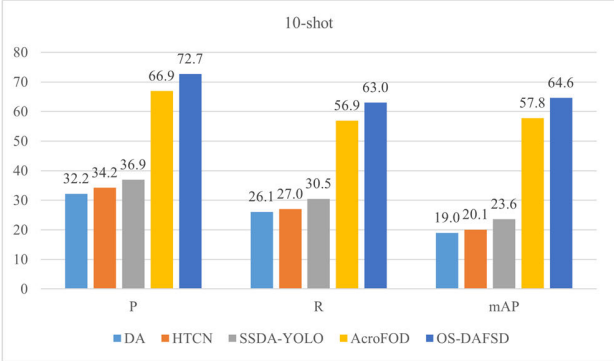


Fig. 10. Graph of Table V.

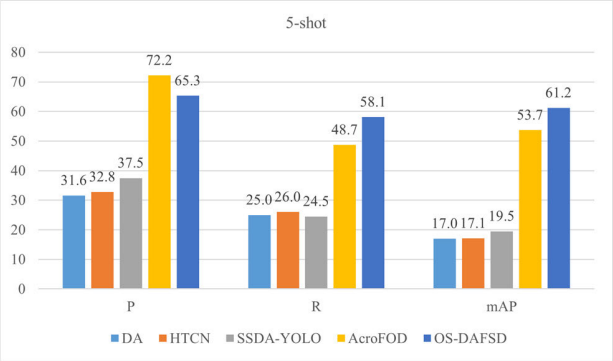


Fig. 9. Graph of Table IV.

on domain-adaptive few-shot SAR image ship detection. Figs. 8–10 show the bar charts comparing the results with several domain adaptation methods on P, R, and mAP, respectively.

In addition to experimentally validating the effectiveness of this article's OS-DAFSD method under three-, five-, and ten-shot settings, three common few-shot settings, we experimentally validate the self-constructed DIOR2SSDD domain-adapted dataset of SAR images of the target domain randomly sampled at the ratio of 10% and 20% from the training set as the few-shot data in the target domain, what performance can be achieved with only 10% and 20% of the target domain data volume, and we compared the performance of this article's method with the performance of classical target detection algorithms [13], [29], [32], [58], [61], [62], [63], [64], [65], [66], in using the full data volume of SSDD, and the experimental results are shown in Table VI.

The specific meaning of the comparison results in Table VI is to demonstrate in our experiment that our proposed method can achieve the detection performance of conventional target detection methods trained on all SAR images by using a small proportion of SAR images from the target domain for training. This is to verify that our proposed method can still maintain excellent performance even in the case of scarce training SAR images and even achieve the performance of conventional target detectors trained on all SAR images with only a small proportion of SAR image training. The experimental results from Table VI demonstrate that our OS-DAFSD can achieve remarkable results with only 10% and 20% of the target domain data volume. It even outperforms some of the

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TABLE IV
COMPARISON RESULTS WITH EXCELLENT DOMAIN-ADAPTIVE TARGET DETECTION METHODS UNDER FIVE-SHOT SETTING

Methods	5-shot ship (%)		
	P	R	mAP
DA	31.6	25.0	17.0
HTCN	32.8	26.0	17.1
SSDA-YOLO	37.5	24.5	19.5
AcroFOD	**72.2**	48.7	53.7
OS-DAFSD	65.3	**58.1**	**61.2**

TABLE V
COMPARISON RESULTS WITH EXCELLENT DOMAIN-ADAPTIVE TARGET DETECTION METHODS UNDER TEN-SHOT SETTING

Methods	10-shot ship (%)		
	P	R	mAP
DA	32.2	26.1	19.0
HTCN	34.2	27.0	20.1
SSDA-YOLO	36.9	30.5	23.6
AcroFOD	66.9	56.9	57.8
OS-DAFSD	**72.7**	**63.0**	**64.6**

TABLE VI
COMPARISON RESULTS WITH TARGET DETECTION METHODS USING ALL DATA

Ratio	Method	ship AP50(%)	mAP50(%)
	RetinaNet	86.2	86.2
	DCMSNN	89.4	89.4
	YOLOX-Nano	90.7	90.7
	SSD300	92.3	92.3
	CornerNet	93.6	93.6
100%	Faster RCNN	93.7	93.7
	FPN	96.5	96.5
	Cascade R-CNN	96.8	96.8
	YOLOX-s	97.2	97.2
	YOLOv5n	97.7	97.7
	HRLE-SARDet	98.4	98.4
10%	**OS-DAFSD**	**89.3**	**89.3**
20%	**OS-DAFSD**	**91.9**	**91.9**

target detection methods that use the full SSDD data volume, such as RetinaNet and YOLOX-Nano. The OS-DAFSD model achieves an mAP of approximately 90.8% compared to the HRLE-SARDet model, which is the current excellent model for SAR image target detection when using only 10% volume of the data. In addition, when using 20% volume of the data, the OS-DAFSD model achieves an mAP of 93.4% compared to the HRLE-SARDet results. These experiments show the effectiveness of the OS-DAFSD in assisting with the task of detecting ship targets in SAR images. Our algorithm makes use of optical image data reasonably, resulting in an effective solution for detecting few-shot ships in SAR images.

2) Ablation Experiments Results of OS-DAFSD on DIOR2SSDD: To assess the effectiveness of each module proposed in the OS-DAFSD algorithm, numerous ablation experiments were conducted on DIOR2SSDD to verify the efficacy of its modules. The sections for the ablation experiments are given as follows.

1) Add the proposed loss branch DKL branch driven by the latent similarity in domain knowledge distribution to the baseline.
2) Based on (1), DFANet is a dual-stream branching instance-level feature alignment extraction network with weight sharing, is used.
3) Based on (2), the distance metric function MMD used in the original baseline is replaced with the proposed efficient plug-and-play distance metric function OSD.

The final results of the ablation experiments are shown in Table VII. Each of the modules proposed in this article brings a certain gain and gives a certain boost to the model's mAP in all three different shot settings. In the three-shot setting, the OSD brings the greatest improvement in mAP with 4.3%, and the DFANet also brings an improvement of 3.9%. In the ten-shot setting, DKL also brings a 1.9% mAP gain. This proves the various modules and strategies we designed.

Besides, t to verify whether the performance of OSD is truly excellent, we compared its performance with MMD and CSD and conducted ablation experiments on the metric function. The experimental results are shown in Table VIII.

The results in Table VIII show that the OSD outperforms the two common metric functions. It provides a better measurement of the distribution distance between the two domains, allowing for improved screening of optical images that closely resemble SAR images. This screening process eliminates harmful training samples and improves the utility of augmented samples, ultimately leading to improved performance.

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TABLE VII
ABLATION RESULTS OF OS-DAFSD ON DIOR2SSDD

Methods	DKL	DFANet	OSD	ship AP50 (%)		
				3-shot	5-shot	10-shot
Baseline	×	×	×	50.5	53.7	57.8
OS-DAFSD	√	×	×	51.0	55.3	59.7
OS-DAFSD	√	√	×	54.9	58.7	61.2
OS-DAFSD	√	√	√	59.2	61.2	64.6

TABLE VIII
ABLATION RESULTS OF METRIC FUNCTION

Methods	MMD	CSD	OSD	ship AP50 (%)		
				3-shot	5-shot	10-shot
Baseline	√	×	×	50.5	55.7	57.8
Baseline	×	√	×	53.7	53.0	57.8
Baseline	×	×	√	**55.5**	**57.3**	**58.3**

3) Visualization:

a) Visualization of data augmentation during the training stage: After selecting optical images with distributions close to SAR images, they are input into the network along with the SAR images for training. During this process, as shown in Fig. 4, we further perform additional data augmentation steps and visualize the augmented images, as shown in Fig. 11. By simultaneously presenting SAR and optical ship images, the model can absorb richer and more diverse information during training, thereby learning more robust features and significantly enhancing its generalization ability.

b) Visualization of qualitative detection results: Fig. 12 displays the typical detection results of the OS-DAFSD model for the DIOR2SSDD dataset, evaluating YOLOv5n, AcroFOD, and our method OS-DAFSD. The red circles are used to mark falsely detected ships and the yellow circles are used to mark missed detected ships. The results show that the YOLOv5n, trained by directly mixing only optical images and few-shot SAR images, does not accurately localize ship targets. Although the domain-adaptive algorithm AcroFOD predicts the bounding boxes of some ship targets more accurately, it also suffers from some omission and false detection problems. In contrast, the model OS-DAFSD in this article successfully suppresses the vast majority of FPs, has fewer missed ships, and accurately localizes ships even when ship targets are near the shore.

c) t-SNE feature distribution visualization: The t-SNE tool is used to visualize the feature distributions of the two domains of the DIOR2SSDD-passed model when using 10% and 20% of the target-domain training data in Fig. 13(a)–(c), where YOLOv5n, AcroFOD, and our method OS-DAFSD are visualized. Compared to the model YOLOv5n, the domain adaptation algorithm AcroFOD, the OS-DAFSD-passed ship features of the two domains are better aligned. Besides, the distributions are more overlappingly blended between a limited number of ships of the target domain (dark orange dots) and a sufficiently large number of ships of the source (blue dots) that many of the source instances will be pushed closer to the target instances. This visually validates that the OS-DAFSD improves the ability to align the two different feature distributions and the ability to filter the source domain data.

d) Feature heatmap visualization: Fig. 14 displays the feature heatmap of the last layer of some test SAR images before the output of the OS-DAFSD model using grad-cam. The heatmap indicates that the OS-DAFSD model has a high feature focus on the ship targets of the test SAR images, enabling effective detection of the ship targets in the sea clutter and complex near-shore background. The OS-DAFSD model is shown to acquire domain knowledge from ample source domain data and successfully adapt to SAR images in the target domain.

V. DISCUSSION

In this section, we further discussed the process of domain adaptation, what knowledge can SAR images transfer from optical images, the limitations of the method proposed in this article, and future work.

A. What Knowledge Transferred From Optical Images

For target domain SAR images, it is important to understand what knowledge is mainly transferred from source domain optical images. We have also conducted relevant analysis through this research work and believe that the transferred knowledge may mainly include the following aspects.

1) *Shape and Structural Characteristics:* Despite notable differences in imaging principles, colors, and textures between optical and SAR images, the basic shape and structure of ships are usually similar in the two images. Therefore, these shape and structural features can be transferred from the optical domain to the SAR domain through transfer learning.
2) *Context Information:* In ship detection, context information is very important. For example, ships typically appear on the water surface and may be surrounded by other ships or marine environments. This contextual information can be learned from optical images and

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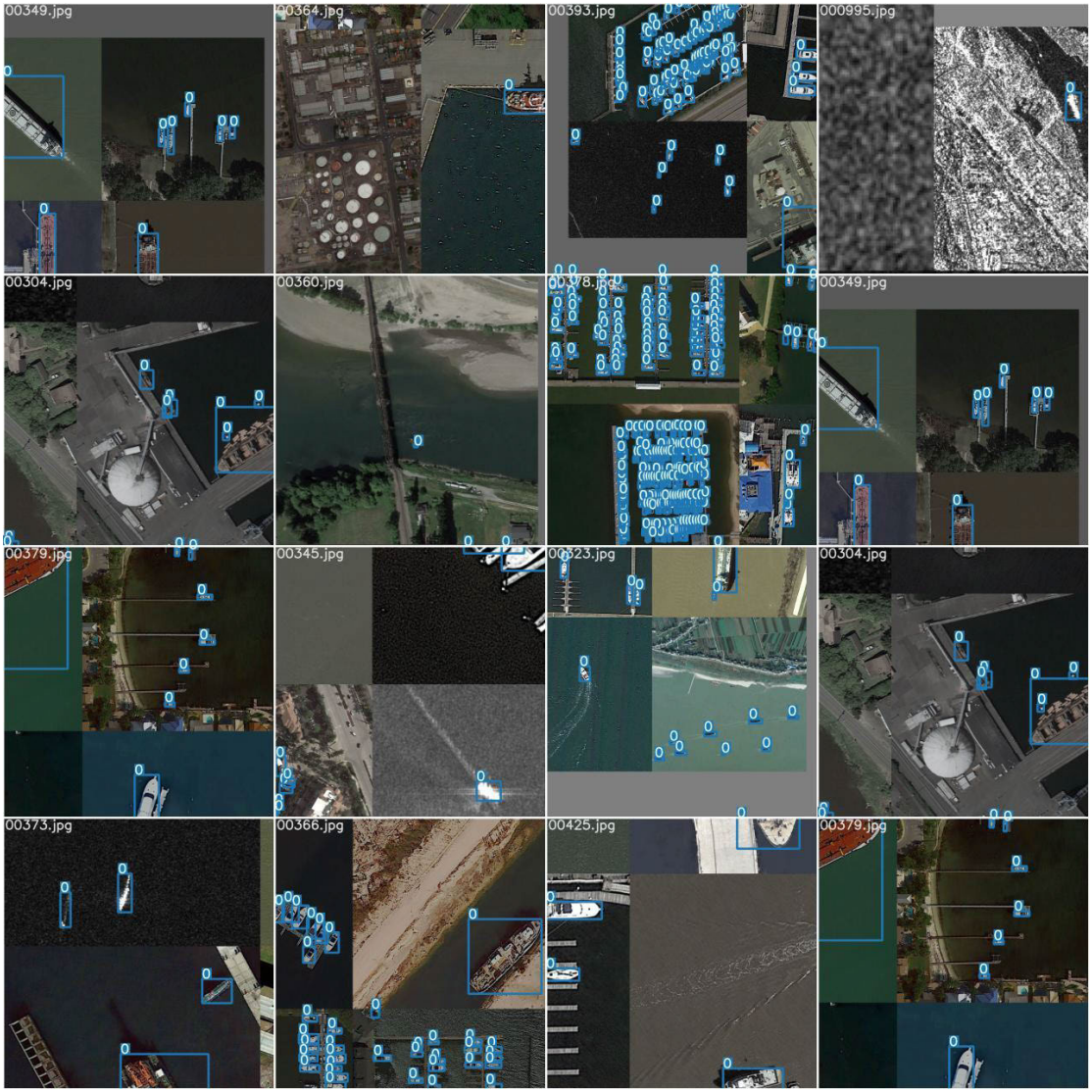


Fig. 11. Visualization of data augmentation during the training stage.

transferred to SAR images to help improve the accuracy of ship detection.

3) *Scale Variation:* Ships may exhibit varying scales in optical and SAR images. Through transfer learning, the capability to detect ships across varying scales can be acquired, thereby enhancing the robustness of cross-domain detection.
4) *Rotation and Perspective Changes:* Ships may undergo rotation and perspective changes in SAR images due to factors such as imaging angle and platform motion. Through domain adaptation, robustness to these changes can be learned from optical domain and applied to the SAR domain.
5) *Feature Extraction Capability:* Convolutional neural networks (CNNs) can learn the mapping from raw pixels to advanced features. Through transfer learning, these feature extraction capabilities can be learned from the optical domain to the SAR domain, thereby improving the ship detection performance in SAR images.

B. Limitation

Several key limitations arise in domain-adaptive few-shot SSD algorithm from optical to SAR images, requiring innovative strategies and solutions. These limitations include:

1) *Data Difference:* There are significant differences in imaging principles, data representation, and characteristics between optical and SAR images. Consequently, models trained on optical images may exhibit reduced performance on SAR images, owing to the divergence in data distribution and feature space between the two image types.
2) *Data Scarcity:* The unique imaging methods and limitations associated with SAR images result in a scarcity of labeled data for specific target categories, particularly in few-shot scenarios. This scarcity poses challenges for the model to learn adequate feature representations, consequently impacting the accuracy of target detection.
3) *Target Diversity:* SAR targets may have different shapes, sizes, and directions, making target detection more complex. In addition, the background in SAR images may

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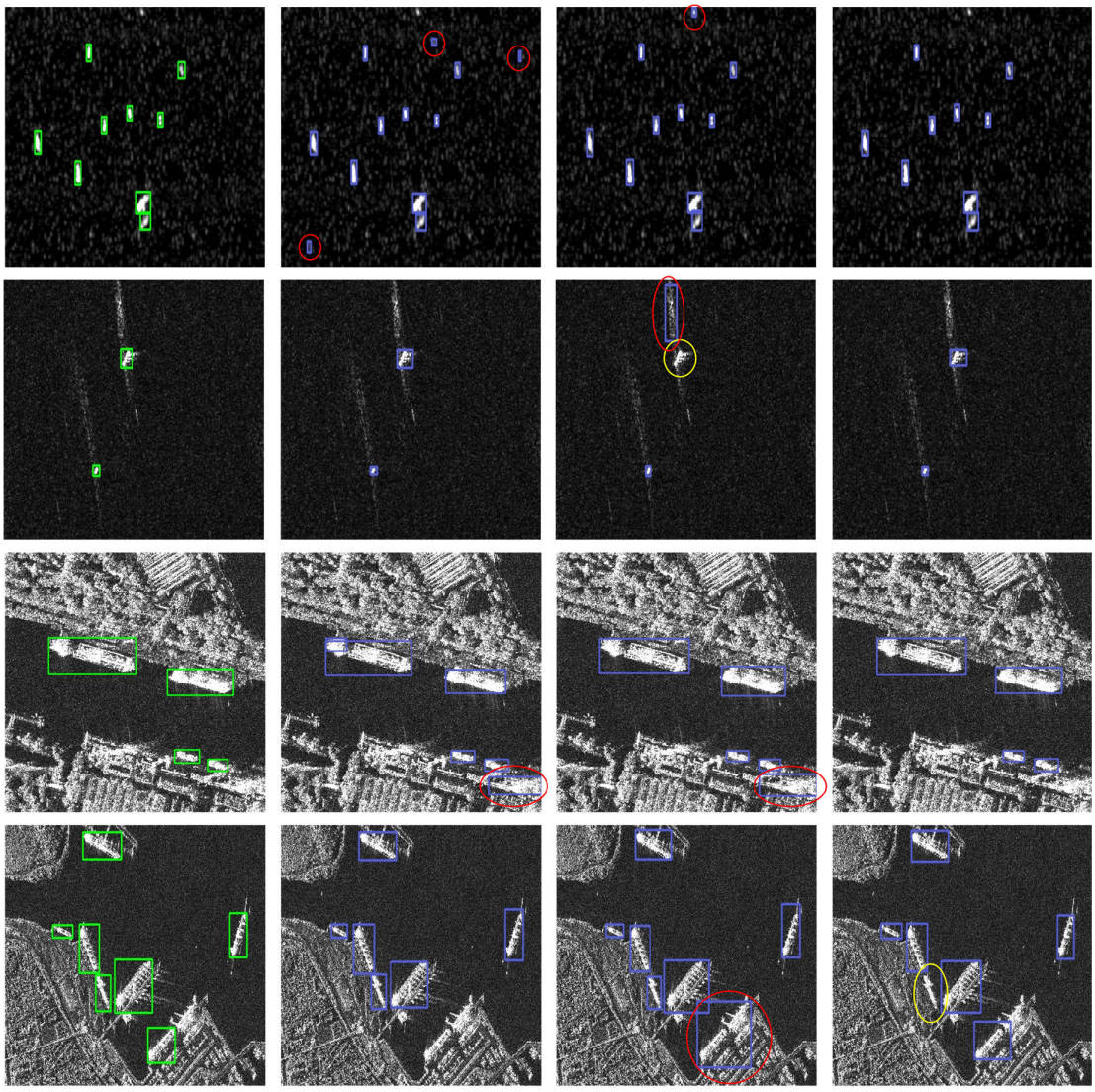


Fig. 12. Visualization of detection results. The red circles are used to mark falsely detected ships and the yellow circles are used to mark missed detected ships. (a) GT. (b) YOLOv5n. (c) AcroFOD. (d) OS-DAFSD.

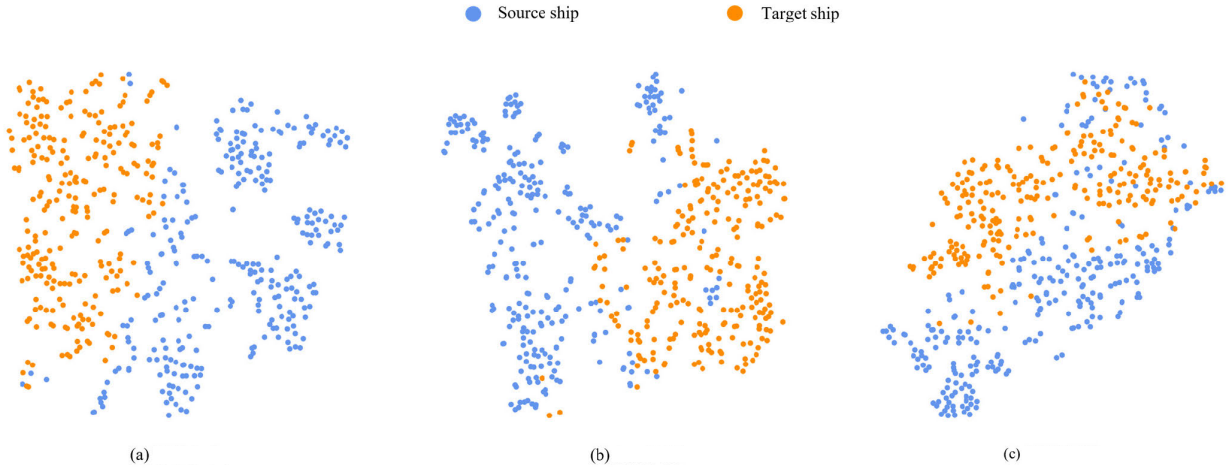


Fig. 13. Feature visualization by t-SNE. (a) YOLOv5n. (b) AcroFOD. (c) OS-DAFSD.

also contain various interference factors, such as terrain, vegetation, and buildings, which further increases the difficulty of target detection.

C. Future Work

To further advance the capabilities of domain-adaptive few-shot SSD algorithm and overcome the existing limitations and

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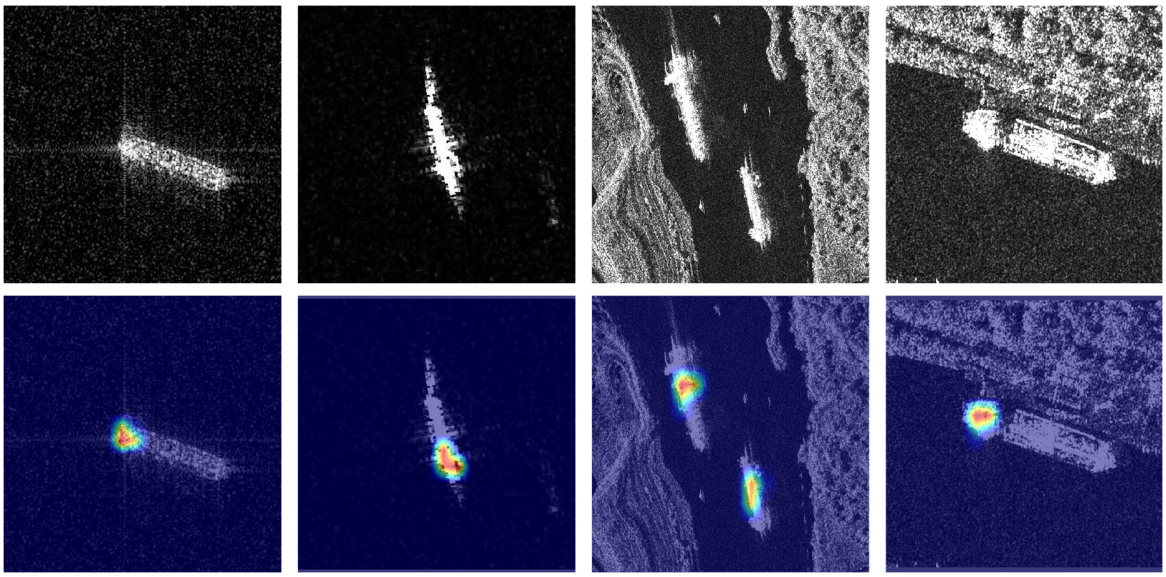


Fig. 14. Heatmap visualization by grad-cam. The first row displays the SAR images of inshore and offshore in actual testing scenarios, and the second row is a visualization diagram of the heat map of the last layer in front of the OS-DAFSD detection head.

challenges, several promising avenues of future research and development are worth exploring, including:

1) *Data Augmentation and Generation:* In response to the issue of data scarcity, future research can focus on developing data augmentation and generation techniques to generate more SAR image data, thereby enhancing the effectiveness of model training. For instance, artificial intelligence generated content (AIGC) or data augmentation techniques can be utilized to generate realistic SAR images.
2) *Multisource Data Fusion:* To fully utilize data from different sources, future research can explore methods for multisource data fusion. By fusing optical images, SAR images, and other types of sensor images (infrared images, hyperspectral images, and so on), richer feature information can be extracted, thereby enhancing the accuracy and robustness of target detection.
3) *Context Information Utilization:* To address the problem of complex background interference in SAR images, future research can explore methods for utilizing context information. By introducing contextual information of target, such as the surrounding environment and relationships between targets, the accuracy or robustness of target detection can be enhanced. For example, graph-based or deep-learning methods can be used to model and utilize contextual information.

VI. Conclusion

This article proposes a domain-adaptive few-shot SAR ship detection algorithm driven by the latent similarity between optical and SAR images. By focusing on data enhancement and feature alignment, it utilizes sufficient and easily accessible optical images to assist ship target detection in SAR images under few-shot conditions and learns from the difference in the distribution of domain knowledge to achieve more accurate migration of domain knowledge between the optical and the SAR image domains. Experiments conducted on the DIOR2SSDD dataset, a domain-adaptive ship target detection dataset transitioning from optical to SAR images, demonstrate the effectiveness of the proposed method in detecting ships in SAR images under few-shot conditions. Furthermore, the experimental results also demonstrate that the method is robust to noise and illumination variations, making it a promising approach for domain-adaptive few-shot SAR image ship target detection. Moreover, our method overcomes the limitation of traditional multisource fusion learning, which requires paired data and enables cross-domain learning for unpaired data. To address the issue of model generalization, we effectively improve the SSD performance through domain-adaptive cross-domain learning.

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